SKYVAULT briefing — tactical_reference_001 v0.1

Summary

Scenario tactical_reference_001 v0.1: 97 events over 10 ticks.

- Result: RED_survived
- Verdict: risky (confidence 0.92)
- Score: 0.50
- Profile: balanced, seen from RED

What happened

Each tick: what was done, and who was destroyed by whom.

- Tick 1: 6 moves
- Tick 2: 6 moves
- Tick 3: 6 moves
- Tick 4: 1 miss; 5 moves
- Tick 5: 4 attacks; 2 misses
- Tick 6: 4 attacks; 1 move; Blue Commander destroyed by Red Rifleman; Red Commander destroyed by Blue Support
- Tick 7: 1 attack; 2 moves; Blue Support destroyed by Red Support
- Tick 8: 2 attacks; 1 miss
- Tick 9: 1 attack; 2 misses
- Tick 10: 2 attacks; Blue Rifleman destroyed by Red Support

Losses

- RED lost 1 of 3: Red Commander at tick 6
- BLUE lost 3 of 3: Blue Commander at tick 6, Blue Support at tick 7, Blue Rifleman at tick 10

Evaluation

Scenario ended because: termination_condition_met

Events recorded: 97

Destroyed entities: 4

Overall score 0.50 under profile balanced: risky (confidence 0.92)

Strongest axis: mission_success 1.00

Weakest axis: recovery_ability 0.00

- mission_success 1.00
- tactical_value 1.00
- strategic_value 0.67
- resource_cost 0.59

- time_cost 0.00
- risk_growth 0.67
- failure_resistance 0.60
- recovery_ability 0.00
- system_robustness 1.00
- stability_impact 0.33
- Failure cause: red_003 destroyed at tick 6 by blue_002
- Failure cause: time_cost 0.00: ended at tick 10 of 10
- Failure cause: recovery_ability 0.00: 0 hp restored against 40 hp still lost
- Failure cause: Anti-gaming: sacrifice_long_term_stability_for_short_term_success
- Next test: Exercise logistics_burden: the world holds no resources to spend
- Next test: Check whether sacrifice_long_term_stability_for_short_term_success was intended: prevent fast success that destroys long-term order
- Flagged: sacrifice_long_term_stability_for_short_term_success

Trade-offs

- tactical_value 1.00 against strategic_value 0.67: both strong
- mission_success 1.00 against resource_cost 0.59: balanced
- time_cost 0.00 against risk_growth 0.67: risk_growth kept at the expense of time_cost
- mission_success 1.00 against stability_impact 0.33: mission_success bought at the expense of stability_impact

Failure exploration

v0.1 stressed 10 times per level; a level breaks below 50% survival.

- Survives without stress: 60%
- Crushed by: communication_failure (assumption actor_policy)
- Best alternative: v0.1-support-rear
- First loss on average at tick 6.1
- sensor_failure: holds at every tested level
- communication_failure: holds up to severe, breaks at collapse
- weather_degradation: holds at every tested level
- enemy_reaction_speed: holds up to moderate, breaks at severe
- equipment_overheat: holds up to severe, breaks at collapse
- civilian_pressure: holds up to severe, breaks at collapse

Variable	mild	moderate	severe	collapse	Breaks at	Sensitivity
sensor_failure	60%	70%	70%	60%	never	+0.00
communication_failure	70%	80%	50%	0%	collapse	+0.60
weather_degradation	70%	80%	100%	60%	never	+0.00

Variable	mild	moderate	severe	collapse	Breaks at	Sensitivity
enemy_reaction_speed	50%	50%	40%	20%	severe	+0.40
equipment_overheat	60%	70%	70%	0%	collapse	+0.60
civilian_pressure	80%	70%	70%	10%	collapse	+0.50

Plan comparison

Batch tactical_reference_001_doctrine: 100 runs per variant, 900 runs, seen from RED.

Conflicting orders: P1 vs P2 (objective, priority, risk); P1 vs P3 (location). Highest win_rate: branch P1 (0.12). Lowest failure_frequency: branch P2+P3 (0.19). Lowest mean_hp_lost: branch P1 (65.98). Lowest mean_risk_events: branch P1 (1.73). Highest mean_score: branch P1 (0.4128). SKYVAULT does not choose between the branches; NEXUS or the user decides which orders to keep.

- Best win rate: support-rear
- Best failure frequency: support-rear
- Best mean hp lost: support-rear
- Best mean risk events: support-rear
- Best mean score: support-rear
- Conflict C1: P1 vs P2 at tick 1: both order red_002; both high priority; risk medium vs low
- Conflict C2: P1 vs P3 at tick 1: both send a unit to [1, 1]
- Decision required: yes

Variant	Win	Fail	HP lost	Risk	Score
plan	6%	27%	69.05	1.92	0.395
support-rear	17%	17%	61.79	1.61	0.434
spread	12%	21%	67.96	1.83	0.415
sweep:entities[blue_002].capabilities.accuracy=0.55	6%	27%	69.05	1.92	0.395
sweep:entities[blue_002].capabilities.accuracy=0.65	7%	38%	74.6	2.11	0.362
sweep:entities[blue_002].capabilities.accuracy=0.75	3%	41%	79.21	2.22	0.343
branch:P1	12%	23%	65.98	1.73	0.413
branch:P2+P3	3%	19%	68.43	1.83	0.404
branch:none	6%	27%	69.05	1.92	0.395

Reproducibility

- Run: run_20261108T044000539618Z_5e71d369
- Random: seed 42 (fixed)
- Runtime 0.13.0, rules rules-2438b4a50419
- Input hash: 5e71d369a6d87f35faf23d1c6f3289da60bede8b1a43de0d843402f6420fd14e
- Output hash: 63090026e13734f596339fe2129a4d17cbd43499f72c87cb0dbec5cdf9f5fd09